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6 October 2026

The Editors
The Journal of Chemical Physics

Submission of a Regular Article: “Closed-Form Storage Stability, Induction Delay, and Thermal-Feedback Criticality of Melt-Gated Autocatalytic Cure”

Dear Editors,

I submit the enclosed theoretical paper for consideration in The Journal of Chemical Physics, in the section on Polymers and Soft Matter (the Theoretical Methods and Algorithms section would also be appropriate). It treats a problem of reaction dynamics: a cure that is triggered by the melting of an encapsulated catalyst, with a distribution of capsule melting temperatures, autocatalytic kinetics and exothermic feedback on the temperature. The central result is that this lumped model is largely solvable in closed form.

What the paper establishes

- At constant temperature the gated cure is exactly the isothermal cure evaluated on the clock of integrated catalyst availability, so the gate delays every conversion level by essentially the release time constant (Proposition 1).
- The isothermal cure time is exact by one quadrature for any reaction order and in closed form for first order; sharpness obeys a logarithmic law in the rate-constant ratio (Proposition 2).
- The storage conversion of a Gaussian capsule population has a closed form in the open fraction $\Phi(-\Delta/\sigma_{\text{eff}})$, and inverts into a design rule for the largest admissible spread of melting temperatures that agrees with the population equations within 3 % (Proposition 3).
- Thermal feedback has a critical number $\Pi_c = \psi\Theta = e^{-1}(1+n)^{(1+n)}/n^n$ at a fold of the quasi-steady manifold, with overshoot given by the Lambert W function, a square-root normal form, and an Arrhenius correction $\exp(1/Ar)$ (Proposition 4).
- In a continuous-flow reactor the same fold gives an ignition–extinction hysteresis with a computed existence threshold, and the loop area extrapolates to the static value as the sweep rate vanishes (Proposition 5).

Every result is tested against the full equations. The paper also states the status and failure modes of each claim, and its falsifiable signatures.

Scope and honesty about it

The paper is purely theoretical and no experimental data are used; all parameters are illustrative and are not fitted to any material. I have tried to state this plainly throughout, including in a table of the status of every claim. The parameter set and the capsule-population picture were introduced in a numerical study that I posted as a preprint (ChemRxiv 10.26434/chemrxiv.15008366; Zenodo 10.5281/zenodo.22073390);

the analytic results here are new, and the comparisons of that study are not used. That preprint is cited in the manuscript.

Data, code and disclosures

All code, tests and figure scripts are public at <https://github.com/sandlerleon/gated-autocatalysis-criticality> and archived at <https://doi.org/10.5281/zenodo.23201763>; the manuscript is available as a preprint at <https://doi.org/10.5281/zenodo.23201767>. A large language model (Claude, Anthropic) assisted with derivations, code, figures and drafting; I reviewed and verified the content, checked every closed-form result against the numerical model, resolved every journal reference through Crossref, and take full responsibility, as stated in the manuscript. I am the sole author, have no competing interests and received no funding. The manuscript is not under consideration elsewhere.

Thank you for considering the paper.

Yours sincerely,

Leon Sandler